

Lowell National Historical Park
NATIONAL PARK SERVICE
PMIS NO. 337427A & C

**Replace Cooling Tower #1, Condenser and Circulator Pumps, and
Associated Valves at Boott Cotton Mill #6**

Scope of Work

This work is for a cooling tower replacement (PMIS 337427A) on the roof of the fifth floor of the Boott Cotton Mill #6, in downtown Lowell, MA. The project involves removing the existing rooftop cooling tower by crane (placed at the adjacent Bridge Street) and installing a new low-rise cooling tower on new vibration rails. Piping connections are to be made at the roof level and replace degraded valves. This is one of two cooling towers for the building.

Work also includes (PMIS 337427C) replacing one 384 gpm 90' head condenser pump and one 500 gpm 70' head circulatory pump (that connects the chiller #1 (PMIS 337240C- Not in contract) to the cooling tower #1 in the basement mechanical room. Work will include adding variable frequency drives to the pumps. Work includes replacing degraded 4 two-way control valves, 16 ball valves, 1 automatic control valve, 4 butterfly valves, approximately 16 lf of pipe and connections necessary to complete the system, 3 sets of vibration springs and mounting to concrete pads; and program the cooling tower and the pumps to the Trane Building Management System. Electrical work shall include any new conduit and wiring. Please refer to the Contract Specifications and Plans for complete Scope.

Option #1 is to replace one circulatory pump in the adjacent building basement mechanical room. Work will include adding variable frequency drive to the pump. Work includes replacing degraded 2 two-way control valves, 8 ball valves, 1 automatic control valve, 2 butterfly valves, approximately 8 lf of pipe and connections necessary to complete the system, 1 set of vibration springs and mounting to existing concrete pad; and program the cooling tower and the pump to the Trane Building Management System. Electrical work shall include any new conduit and wiring. Please refer to the Contract Specifications and Plans for complete Scope.

